

adiabatic
 $p = w\rho = (1 + \frac{1}{3})\rho$
 $C_s = \left(\frac{\partial p}{\partial \rho}\right)_s$

$$\Gamma = \frac{C_p}{C_v}$$

Jeeun: homogeneous, self-gravitating medium

$$(1) \quad \nabla^2 \phi = 4\pi G \rho$$

$$(2) \quad (\partial_t + \vec{v} \cdot \vec{\nabla}) \vec{v} = -\frac{C_s}{\rho} \vec{\nabla} \rho - \vec{\nabla} \phi$$

Steady solution $\rho_0(\vec{r}, t) = \rho_0$

$\vec{v}_0(\vec{r}, t) = 0$ (stationary medium)

$\phi_0(\vec{r}, t) = \dots$

perturbation $\phi(\vec{r}, t) = \phi_0(\vec{r}) + \phi_1(\vec{r}, t)$

$\rho(\vec{r}, t) = \rho_0 + \rho_1(\vec{r}, t)$

$\vec{v}(\vec{r}, t) = \vec{v}_0 + \vec{v}_1(\vec{r}, t)$

$C_s(\vec{r}, t) = C_{s,0} + C_{s,1}(\vec{r}, t)$

$$\rho_1, u_1 \sim O(\varepsilon)$$

continuity:

$$\partial_t \rho + \vec{\nabla} \cdot (\rho \vec{u}) \rightarrow 0$$

$$\vec{u}_0 = 0$$

$$\rho_0 = \text{const}$$

$$\partial_t (\rho_0 + \rho_1) + \vec{\nabla} \cdot ((\rho_0 + \rho_1)(\vec{u}_0 + \vec{u}_1))$$

$$= \cancel{\partial_t \rho_0} + \partial_t \rho_1 + \cancel{\vec{\nabla} \cdot (\rho_0 \vec{u}_0)} +$$

$$+ (\rho_0 + \rho_1) \vec{\nabla} \cdot \vec{u}_1 + \vec{u}_1 \cdot \vec{\nabla} (\rho_0 + \rho_1)$$

$$= \partial_t \rho_1 + (\rho_0 + \rho_1) \vec{\nabla} \cdot \vec{u}_1 + \vec{u}_1 \cdot \vec{\nabla} \rho_1$$

$$= \underbrace{[\partial_t \rho_1 + \rho_0 \vec{\nabla} \cdot \vec{u}_1]}_{O(\varepsilon)} + \underbrace{[\rho_1 \vec{\nabla} \cdot \vec{u}_1 + \vec{u}_1 \cdot \vec{\nabla} \rho_1]}_{O(\varepsilon^2)}$$

Poisson:

$$\nabla^2 (\phi_0 + \phi_1) = 4\pi G (\rho_0 + \rho_1)$$

$$[\cancel{\nabla^2 \phi_0} - 4\pi G \rho_0] + [\nabla^2 \phi_1 - 4\pi G \rho_1] = 0$$

$$\nabla^2 \phi_1 = 4\pi G \rho_1$$

$$\boxed{0 = \frac{d\vec{U}_0}{dt} = -\frac{C_s^2}{\rho} \vec{\nabla} \rho_0 - \vec{\nabla} \phi_0 \Rightarrow \vec{\nabla} \phi_0 = 0}$$

Euler: $\frac{d\vec{U}}{dt} = (\partial_t + \vec{U} \cdot \vec{\nabla}) \vec{U} = -\frac{C_s^2}{\rho} \vec{\nabla} \rho - \vec{\nabla} \phi$

$$\partial_t \vec{U}_1 + \underbrace{(\vec{U}_0 \cdot \vec{\nabla}) \vec{U}_1}_{O(\phi)} = \frac{-(C_{s,0}^2 + C_{s,1}^2 + C_{s,0} C_{s,1})}{\rho_0 + \rho_1} \vec{\nabla} (\rho_0 + \rho_1) + \vec{\nabla} (\phi_0 + \phi_1)$$

$$\partial_t \vec{U}_1 = -\frac{C_{s,0}^2}{\rho_0} \vec{\nabla} \rho_1 - \vec{\nabla} \phi_1$$

grad / $\nabla \cdot \rightarrow$

$$\partial_t (\vec{\nabla} \cdot \vec{U}_1) = -\frac{C_{s,0}^2}{\rho_0} \nabla^2 \rho_1 - \nabla^2 \phi_1$$

Poisson

$$= -\frac{C_{s,0}^2}{\rho_0} \nabla^2 \rho_1 - 4\pi G \rho_1$$

continuity

$$\partial_t \left(-\frac{1}{\rho_0} \partial_t \rho_1 \right)$$

$$\rho_0 + \delta \rho$$

$$\Rightarrow \frac{1}{\rho_0} \frac{\partial^2 \rho_1}{\partial t^2} = \frac{C_{s,0}^2}{\rho_0} \nabla^2 \rho_1 + 4\pi G \rho_1$$

symmetry
 $\Rightarrow$ no $\vec{r}, t$
dependence

solving with Fourier modes

$$\rho_1 \approx \rho_k e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\omega = \pm c_{s,0} K \sqrt{1 - \frac{4\pi G \rho_0}{c_{s,0}^2 K^2}}$$

$$= \pm c_{s,0} K \sqrt{1 - \frac{\lambda}{(c_{s,0}^2 \pi / G \rho_0)}} = \pm c_{s,0} K \sqrt{1 - \frac{\lambda}{\lambda_J}}$$

$$\omega^2 \rho_K - c_{s,0}^2 K^2 \rho_K - 4\pi G \rho_0 \rho_K = 0$$

$$\boxed{\omega^2 = c_{s,0}^2 K^2 - 4\pi G \rho_0} \quad \text{Jeans dispersion relation}$$

$$(\omega = 0) \quad c_{s,0}^2 K^2 = 4\pi G \rho_0 \Rightarrow K = \frac{2\sqrt{\pi G \rho_0}}{c_{s,0}}$$

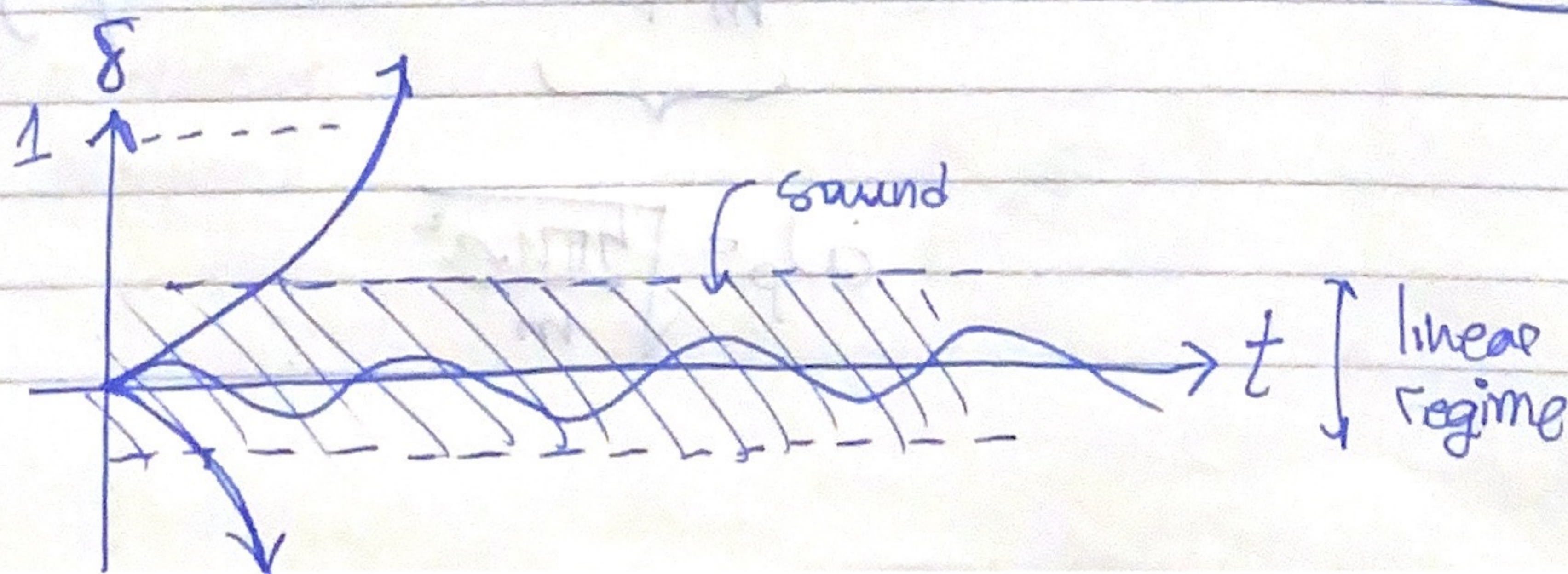
$$K = \frac{2\pi}{\lambda} \Rightarrow \boxed{\lambda_{\text{Jeans}} = c_{s,0} \sqrt{\frac{\pi}{G \rho_0}}}$$

$$\lambda > \lambda_J \Leftrightarrow K < K_J \Leftrightarrow \omega^2 < 0 \quad \text{instability (growing mode)}$$

$$\lambda < \lambda_J \Leftrightarrow K > K_J \Leftrightarrow \omega^2 > 0 \quad \text{stability (oscillatory)}$$

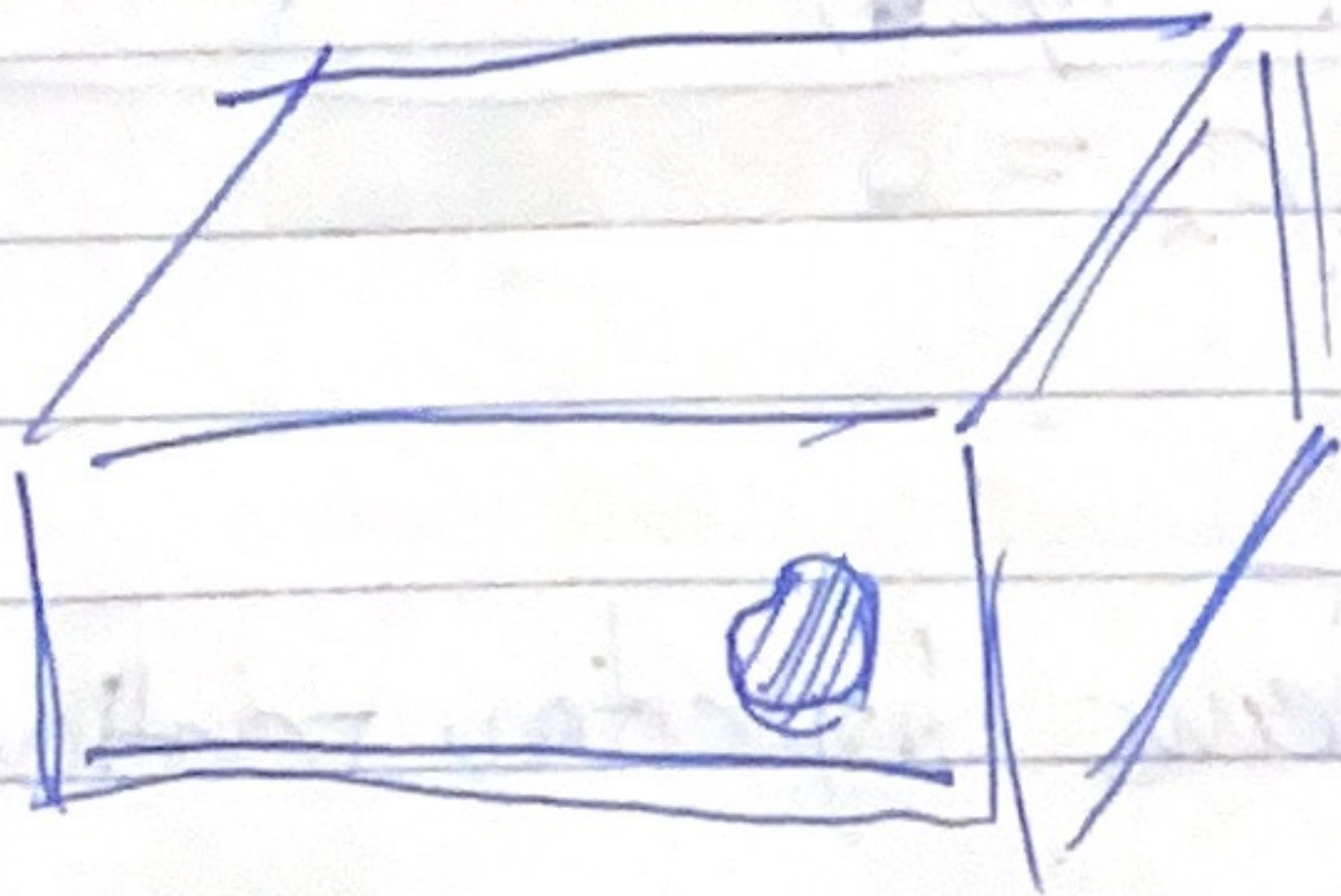
$$\delta = \frac{\rho - \langle \rho \rangle}{\langle \rho \rangle} \rightarrow \frac{(\rho_1 + \rho_0) - \rho_0}{\rho_0} = \frac{\rho_1}{\rho_0}$$

$$\delta \propto e^{i\omega t} \approx \begin{cases} e^{iK c_{s,0} t} & \lambda < \lambda_J \\ e^{\sqrt{4\pi G \rho_0} t} & \lambda > \lambda_J \end{cases}$$



$$\omega = 2\pi\nu$$

Real Space

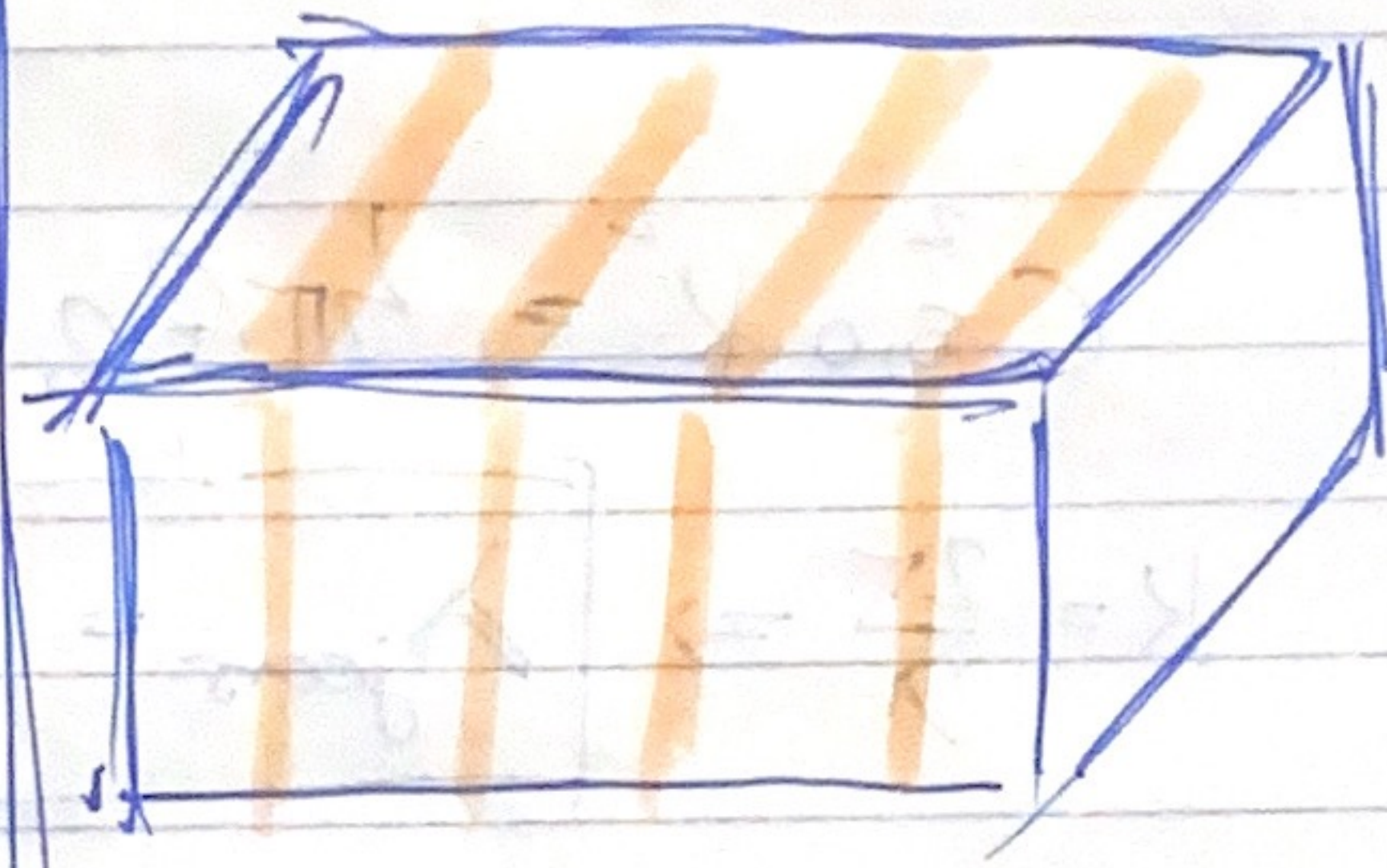
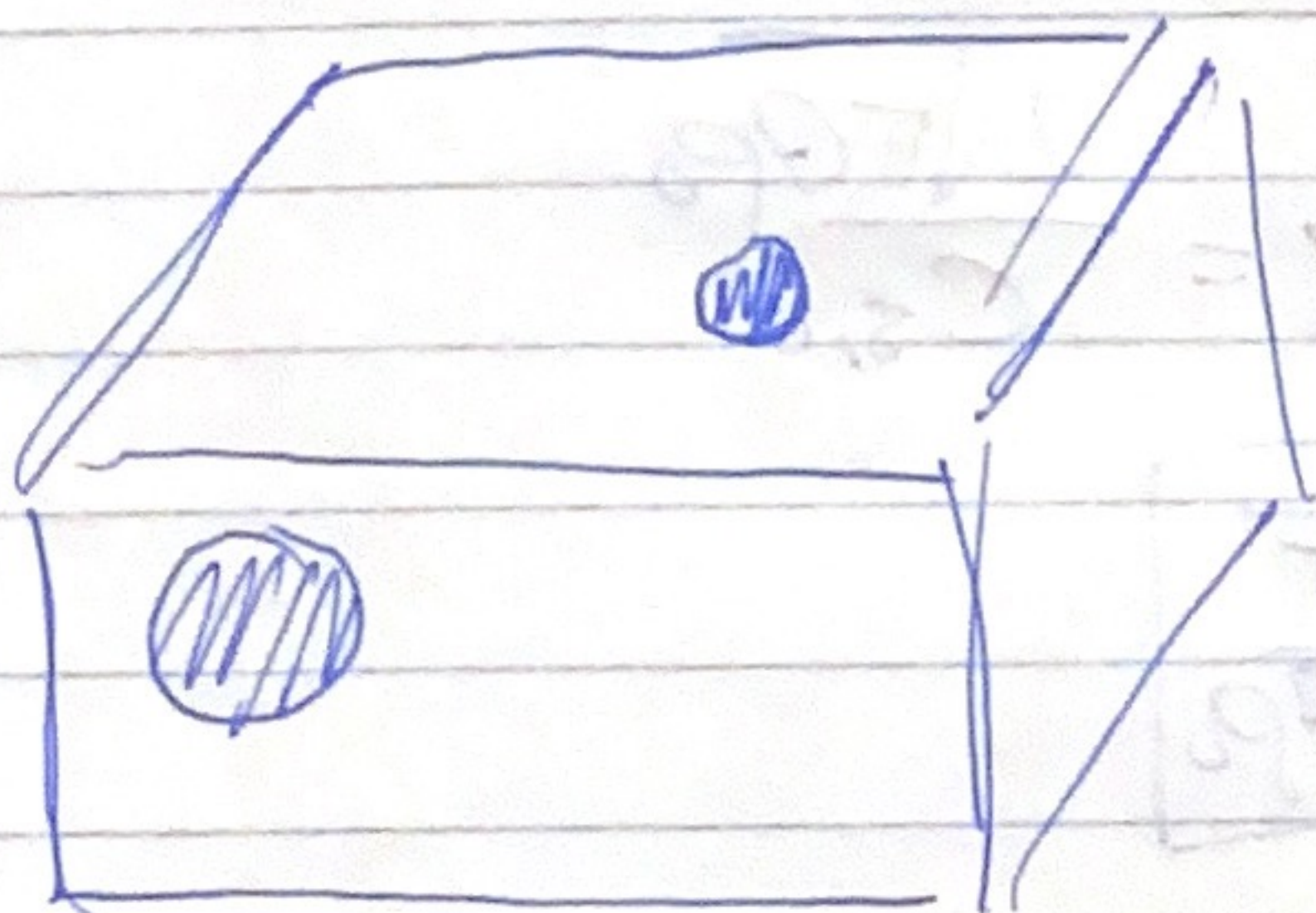


Fourier mode



$K \rightarrow$

$K < K_f \Rightarrow \lambda > \lambda_f$ (long !!)



for $G \rightarrow 0$ $\omega^2 = C_{80}^2 K^2 \Rightarrow \omega = \pm C_{80} K$

Ideal gas: $C_s^2 = \left(\frac{\partial p}{\partial \rho} \right)_s = \frac{\Gamma p}{\rho} = \frac{\Gamma n k_B T}{\bar{m} n} = \frac{\Gamma}{\bar{m}} k_B T$

EM analogy

$G m^2 \rightarrow q^2$ (CGS)
 $C_s \rightarrow c$

$$\left\{ \omega^2 = c^2 K^2 + \frac{4\pi}{m} q^2 n_0 = c^2 K^2 + \omega_p^2 \right\}$$

$$\omega_p = \sqrt{\frac{4\pi n_0 e^2}{m}}$$

$$n_e \approx 10^5 \text{ cm}^{-3}$$

$$\nu \gg \nu_p$$

$$\nu_p = \frac{\omega_p}{2\pi} = \sqrt{\frac{n_e e^2}{\pi m_e}} \approx \sqrt{\frac{10^5}{3 \times 10^{-27}}} \cdot 5 \times 10^{-10}$$

$$\nu_p \sim 3 \times 10^6 \text{ Hz} \approx 3 \text{ MHz}$$

for signals to travel through the ionosphere
the frequency has to be larger than ν_p

Janus Collapse & Fragmentation

$$M_J = \underbrace{\frac{4\pi}{3} \left(\frac{\lambda_J}{2}\right)^3}_{V_J \text{ (sphere)}} \cdot \rho = \frac{\pi}{6} \lambda_J^3 \rho = \frac{\pi}{6} \frac{C_s^3}{(G\rho)^{3/2}} \rho \sim T^{3/2} n^{-1/2}$$

